About 2,600 results for “review wildfire spread prediction”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Predicting climate change effects on wildfires requires linking processes across scales

M. Macias Fauria | S. Michaletz | E. Johnson | Environmental Science · 2011

Accurate process-based **prediction** of climate change effects on **wildfires** requires coupling processes across orders of magnitude of time and space scales, because climate dynamic processes operate at… Expand

74 | PDF | Wiley | Save | Cite

CNN-BiLSTM: A Novel Deep Learning Model for Near-Real-Time Daily Wildfire Spread Prediction

Mohammad Marjani | M. Mahdianpari | F. Mohammadimanesh

Environmental Science, Computer Science · Remote Sensing · 20 April 2024

TLDR This study uses the Visible Infrared Imaging Radiometer Suite (VIIRS) active fire product and a wide range of environmental variables, including topography, land cover, temperature, NDVI, wind informaiton, precipitation, soil moisture, and runoff to train the CNN-BiLSTM model. Expand

38 | [PDF] | PDF | Save | Cite

Wildfire Spread Prediction in North America Using Satellite Imagery and Vision Transformer

Bronte Sihan Li | Ryan Rad | Environmental Science, Computer Science · Conference on Algebraic Informatics · 25 June 2024

TLDR A novel deep learning model, the Attention Swin U-net with Focal Modulation (ASUFM) designed to predict **wildfire spread** in North America using large-scale remote sensing data, offering a promising tool for risk mitigation and resource allocation in **wildfire** management. Expand

3 | PDF | IEEE | Save | Cite

Comparing Accuracy of Wildfire Spread Prediction Models under Different Data Deficiency Conditions

Jiahao Zhou | Wenyu Jiang | Fei Wang | Yuming Qiao | Q. Meng | Environmental Science · Fire · 16 April 2024

Wildfire is one of the most severe natural disasters globally, profoundly affecting natural ecology, economy, and health and safety. Precisely predicting the **spread** of **wildfires** has become an… Expand

1 | PDF | Publisher | Save | Cite

Adaptive wildfire spread prediction for complex terrain: modeling the effectiveness of sprinkler systems

Jaekyoung Kim | Junghyeon Ahn | Junsuk Kang | Environmental Science, Engineering · Fire Ecology · 27 August 2024

Because the threat of **wildfires** to global ecosystems and society continues to rise, this study provides an experimental simulation framework that assesses the **spread** and reduction of **wildfires** to… Expand

4 | PDF | Publisher | Save | Cite

Metaheuristic algorithms for calibration of two-dimensional wildfire spread prediction model

Jorge Pereira · Jérôme Mendes · J. S. Junior · C. Viegas · J. Paulo

Environmental Science, Engineering · Engineering applications of artificial… · 1 October 2024

6 · Publisher · Save · Cite

THE LARGE-SCALE WILDFIRE SPREAD PREDICTION USING A MULTI-KERNEL CONVOLUTIONAL NEURAL NETWORK

M. Marjani · M. Mesgari · Environmental Science, Computer Science ·

ISPRS Annals of the Photogrammetry, Remote… · 14 January 2023

TLDR A multi-kernel convolution neural network (CNN) deep learning model was proposed based on elevation, wind direction, and speed, minimum and maximum temperatures, humidity, precipitation, drought index, normalized difference vegetation index, and energy release component to predict **wildfire spread** across the United States. Expand

23 · [PDF] · PDF · Save · Cite

Wildfire Spread Prediction Through Remote Sensing and UAV Imagery-Driven Machine Learning Models

M. Mohebbi · Elian Wira Sena · Mario Döller · Julian Klinger

Environmental Science, Computer Science · International Conference on Control, Automation… · 12 December 2024

TLDR This paper proposes the utilization of data obtained from Unmanned Aerial Vehicles (UAVs) to develop predictive models for fire **spread**, and AdaBoost and Random Forest demonstrate superior predictive capabilities. Expand

1 · IEEE · Save · Cite

Wildfire spread prediction using geostationary satellite observation data and directional ROS adjustment factor.

Seungmin Yoo · Won-Hee Kang · Junho Song · Environmental Science, Engineering ·

Journal of Environmental Management · 19 November 2024

5 · PubMed · Save · Cite

WildfireSpreadTS: A dataset of multi-modal time series for wildfire spread prediction

Sebastian Gerard · Yu Zhao · Josephine Sullivan · Environmental Science ·

Neural Information Processing Systems · 2023

We present a multi-temporal, multi-modal remote-sensing dataset for predicting how active **wildfires** will **spread** at a resolution of 24 hours. The dataset consists of 13 607 images across 607 fire… Expand

23 · PDF · View Paper · Save · Cite

← 1 2 3 4 … →

Stay Connected With Semantic Scholar

Your E-mail Address · Sign Up

What Is Semantic Scholar?

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

About
About Us
Publishers
Blog

Product
Product Overview
Semantic Reader
Scholar's Hub

API
API Overview
API Tutorials
API Documentation

Research
Publications
Research Careers
Resources

Help
FAQ
Librarians
Tutorials